ARGHADEEP SARKAR

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
Burdwan Municipal High School	Class XII (West Bengal Council of Higher Secondary Education)	2013	79.00%
Burdwan Municipal High School	Class X (West Bengal Board of Secondary Education)	2011	88.25%

ACHIEVEMENTS

Qualified GATE (Graduate Aptitude Test in Engineering) in 2019.

TECHNICAL SKILLS

EDA Tools: Cadence Virtuoso, ADE, Spectre, Calibre/DRC/LVS/PERC, Xilinx Vivado, LTSPICE, MATLAB/Simulink

IC Design: Analog/Mixed-Signal IC Design, Analog Front-End (AFE), Delta Sigma ADC, HV Circuits, Physical Design, Layout

Technology: TSMC 180-nm BCD

Programming: Verilog, Verilog-A/AMS, Python, C

Hardware Platforms: FPGA, Embedded systems

Operating Systems: Windows, Linux

RELEVANT COURSES

1. Digital IC Design
2. Introduction to VLSI Design
3. Embedded System Hardware and Design

IC DESIGN/ VLSI PROJECTS

1. **A cost-effective, real- time, and accurate battery management system (BMS) controller unit for space-qualified Li-ion batteries (August 2021- Present)- Funded by DSP, DRDO.**

- Developed mixed-signal architectures for battery voltage and current sensing.

- Focused on accurate, scalable, and real-time battery pack monitoring.

2. **AI-enabled energy-efficient and lightweight battery management system on chip for advanced energy storage (August 2024- Present)- Funded by TDG.**

- Working on an energy-efficient battery management SoC architecture.

- Contributing to circuit/system design for intelligent monitoring and control.

RESEARCH EXPERIENCE

PhD Researcher, Indian Institute of Technology Hyderabad Aug 2021 - Present

Advisor: Prof. Amit Acharyya

- Designed a mixed-signal sensing front-end architecture for simultaneous series-parallel battery monitoring in TSMC 180nm BCD technology.

- Developed and validated circuits including HV subtractors, analog multiplexers, ADC-interfaced sensing blocks, and mixed-signal control paths.

- Achieved high-precision sensing performance with measurement resolution below 0.1 mV.

- Worked on active cell balancing architectures, switching control, and system-level battery pack monitoring.

- Performed circuit design, simulation, and mixed-signal validation using Cadence Virtuoso and MATLAB/Simulink.

M.Tech (Teaching Assistant), Indian Institute of Engineering Science and Technology (IIEST), Shibpur Aug 2019 - July 2021

Advisor: Prof. Ayan Banerjee

- Worked on Design of an Energy-Efficient Memristive Synapse for Real-Time Realization of spiking neural network-based NeuSoCs.

- Designed energy-efficient circuit blocks for neuromorphic hardware implementation.

PUBLICATION DETAILS

Sarkar, A., Athish, S.A., Dutt, R. and Acharyya, A. 2025, August. A mixed signal sensing front- end architecture for simultaneous series- parallel battery monitoring. In 2025 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS) (pp. 1-5), Lansing, Michigan, USA, 10-13 August. IEEE.
https://doi.org/10.1109/MWSCAS53549.2025.11244574

- Developed a mixed-signal Analog Front-End (AFE) architecture using TSMC 180 nm BCD, integrating HV subtractors, analog multiplexers, and dual 16-bit ADCs for simultaneous series-parallel battery monitoring.
- Achieved high- precision sensing (<0.1 mV) with an 80% improvement in measurement accuracy over state- of- the-art commercial AFE systems.
- Validated complete mixed- signal functionality using Cadence Virtuoso , demonstrating accurate voltage and current sensing across 4S1P-4S4P configurations.

Sarkar, A., Dutt, R. and Acharyya, A., 2025, May. Simultaneous Series- Parallel High- Speed and Accurate Active Cell Balancing for Improved Battery Life. In 2025 IEEE International Symposium on Circuits and Systems (ISCAS) (pp. 1-5). IEEE. *https://doi.org/10.1109/ISCAS56072.2025.11044073*

- Proposed a simultaneous series-parallel active cell balancing methodology achieving higher speed and accuracy for Li-ion packs.
- Validated through MATLAB Simulink for a 4S4P battery pack, showing a reduction in overall power consumption without compromising the balancing accuracy when compared to the state-of-the-art active cell balancing.
- There is also an improvement in balancing speed due to simultaneous energy redistribution among the cells in any string of the battery pack.

Sarkar, A., Dutt, R., Sahu, S., and Acharyya, A., 2023, June. Energy- efficient and High- Speed Active Cell Balancing Methodology for Lithium- ion Battery Pack. In 2023 21st IEEE Interregional NEWCAS Conference (NEWCAS) (pp. 1-5). IEEE. *https://doi.org/10.1109/NEWCAS57931.2023.10198124*

- Presented an energy-efficient and high-speed active cell balancing methodology for series and parallel-connected Li-ion battery packs.
- Designed a control strategy that minimizes energy loss while accelerating equalization between cells.
- Demonstrated significant improvement in balancing speed and efficiency through MATLAB Simulink on various series and parallel-connected battery packs.

Dutt, R., Sahu, S., Sarkar, A., and Acharyya, A., 2023, June. Next- Generation Battery Management System Design Methodology. In 2023 21st IEEE Interregional NEWCAS Conference (NEWCAS) (pp. 1-2). IEEE.
https://doi.org/10.1109/NEWCAS57931.2023.10198048

- Proposed a next-generation design methodology for advanced battery management systems.

- Integrated accurate state estimation, active cell balancing, and efficient control strategies into a unified framework.
- Highlighted the scalability and applicability of the methodology for emerging energy storage and electric vehicle systems.

PATENTS

A METHOD AND A SYSTEM FOR PERFORMING AN ACTIVE CELL BALANCING IN A BATTERY ENERGY STORAGE SYSTEM

Indian Patent No.: 589534

Filing Date: 10/06/2025

Status: Granted

Date of Grant: 15/05/2026

- Designed an active cell balancing methodology that detects SOC imbalance beyond a predefined threshold and triggers mixed-signal control actions for battery cell equalization.
- Implemented a cell-grouping algorithm to pair high-SOC and low-SOC cells, enabling efficient charge redistribution across series-parallel cell configurations.
- Developed MOSFET-based switching paths with precise control logic to enable safe current flow between cells.
- Achieved simultaneous active cell balancing across multi-cell topologies, improving pack uniformity and energy efficiency.

AREA OF INTEREST

Battery Management Systems, Active Cell Balancing Methodology, Lithium-ion Batteries, Flyback Converter, Digital IC Design, VLSI Physical Design, Mixed-signal SoC Design

EXTRACURRICULAR

1. Listening to music

2. Playing Badminton, Squash, and Table Tennis

PERSONAL DETAILS

Date of Birth: 26/12/1995

Sex: Male

Nationality: Indian

Marital Status: Unmarried

Languages Known: English, Hindi, and Bengali.

Permanent Address: Kachipukur North, Chandmari 2nd by lane, Burdwan, East Bardhaman, West Bengal-713101, India.